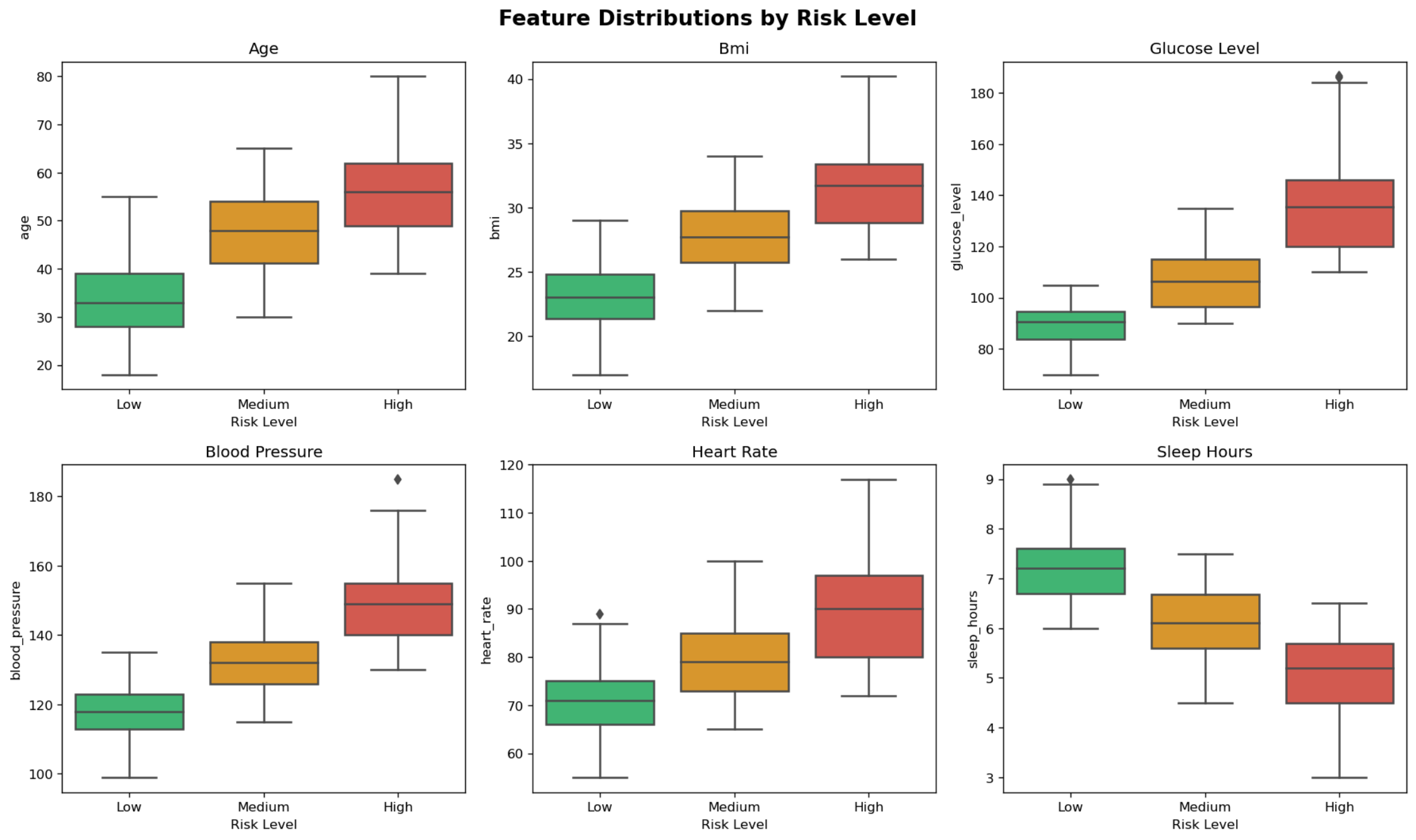
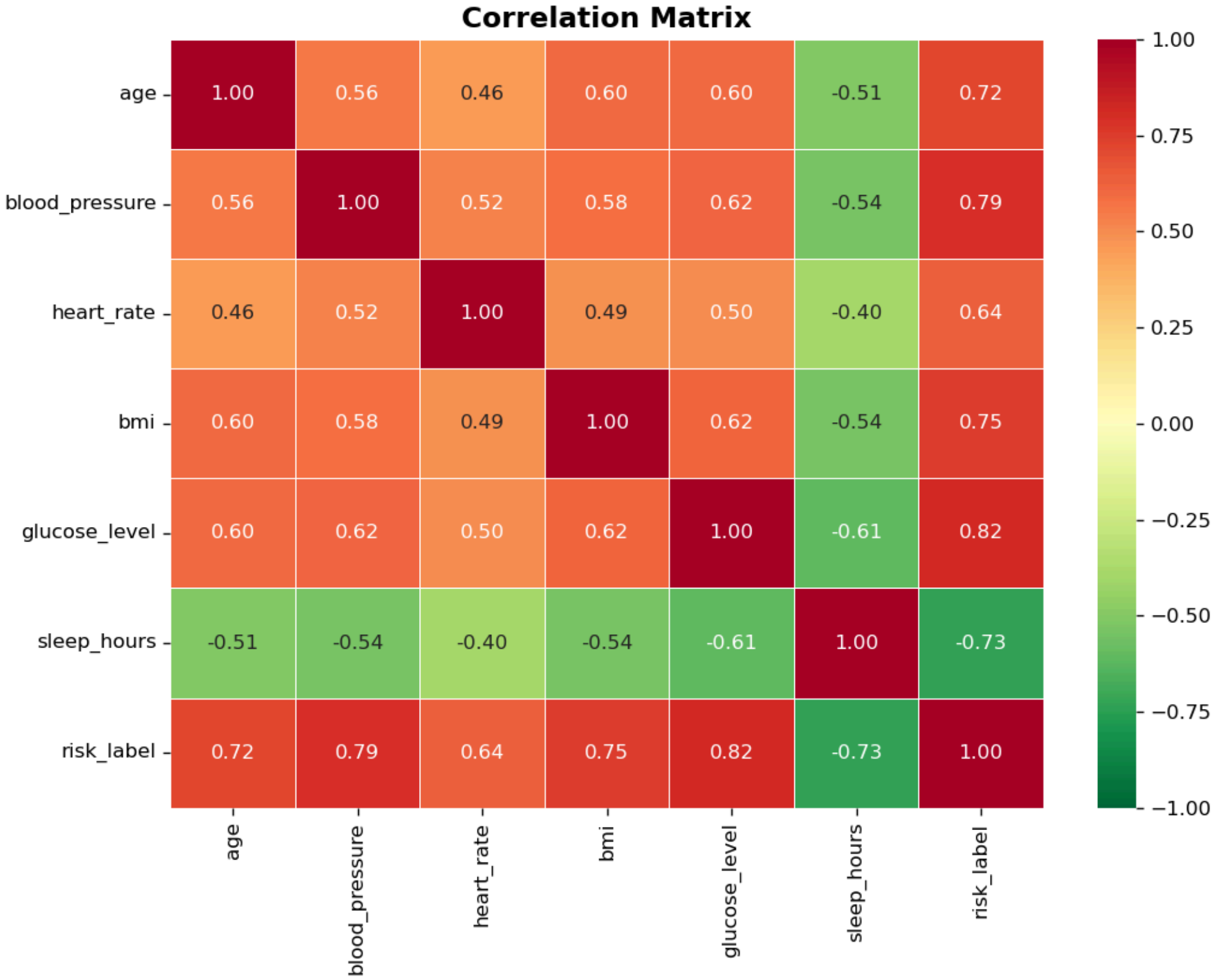
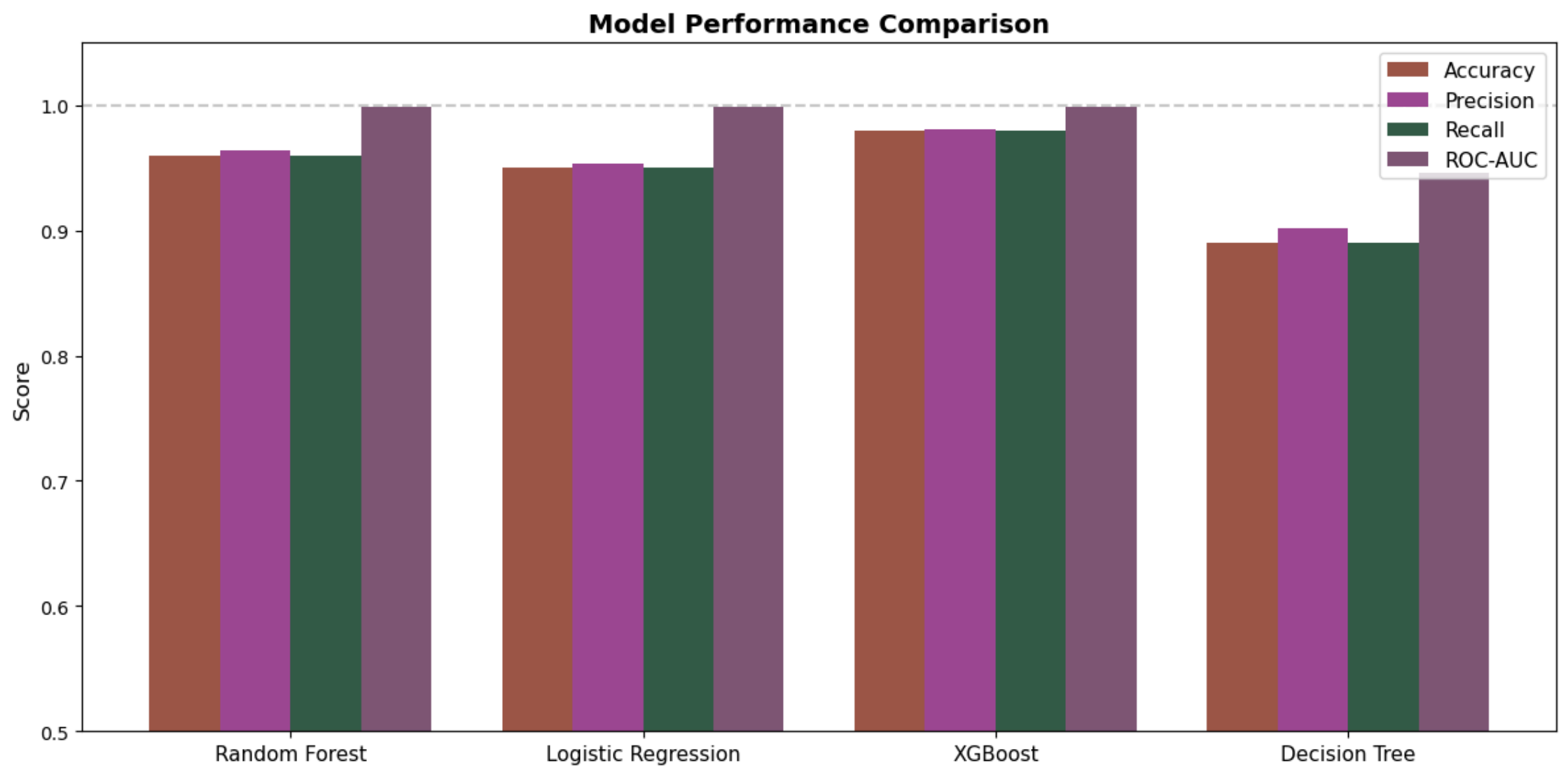
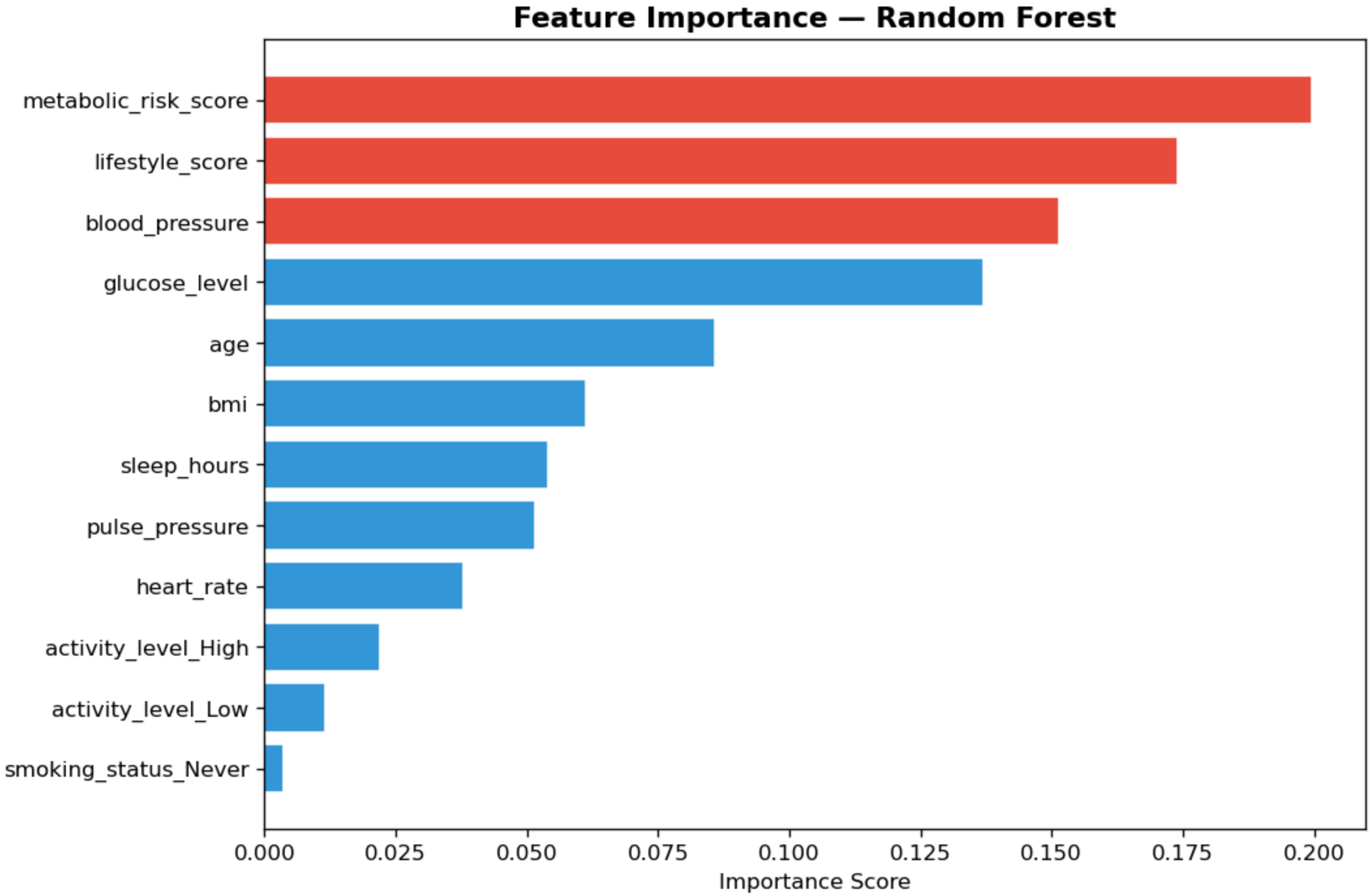
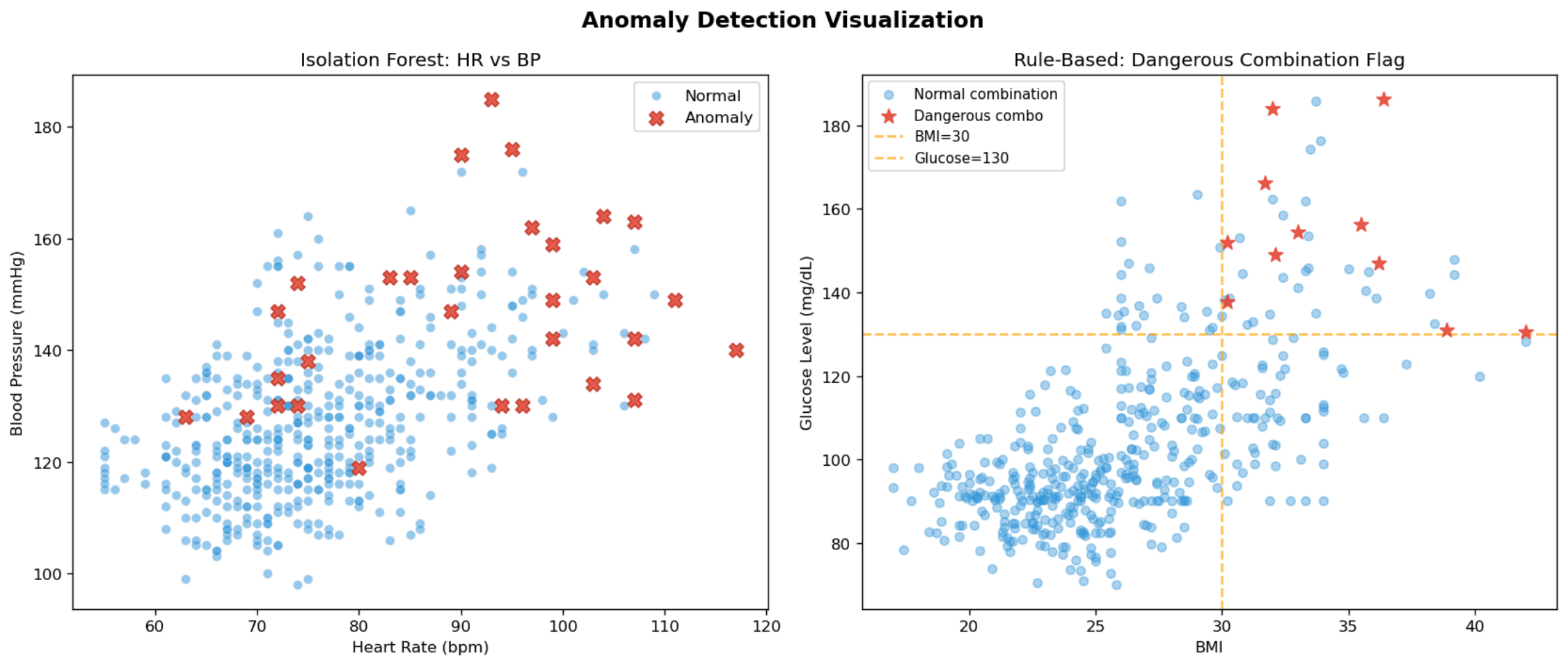
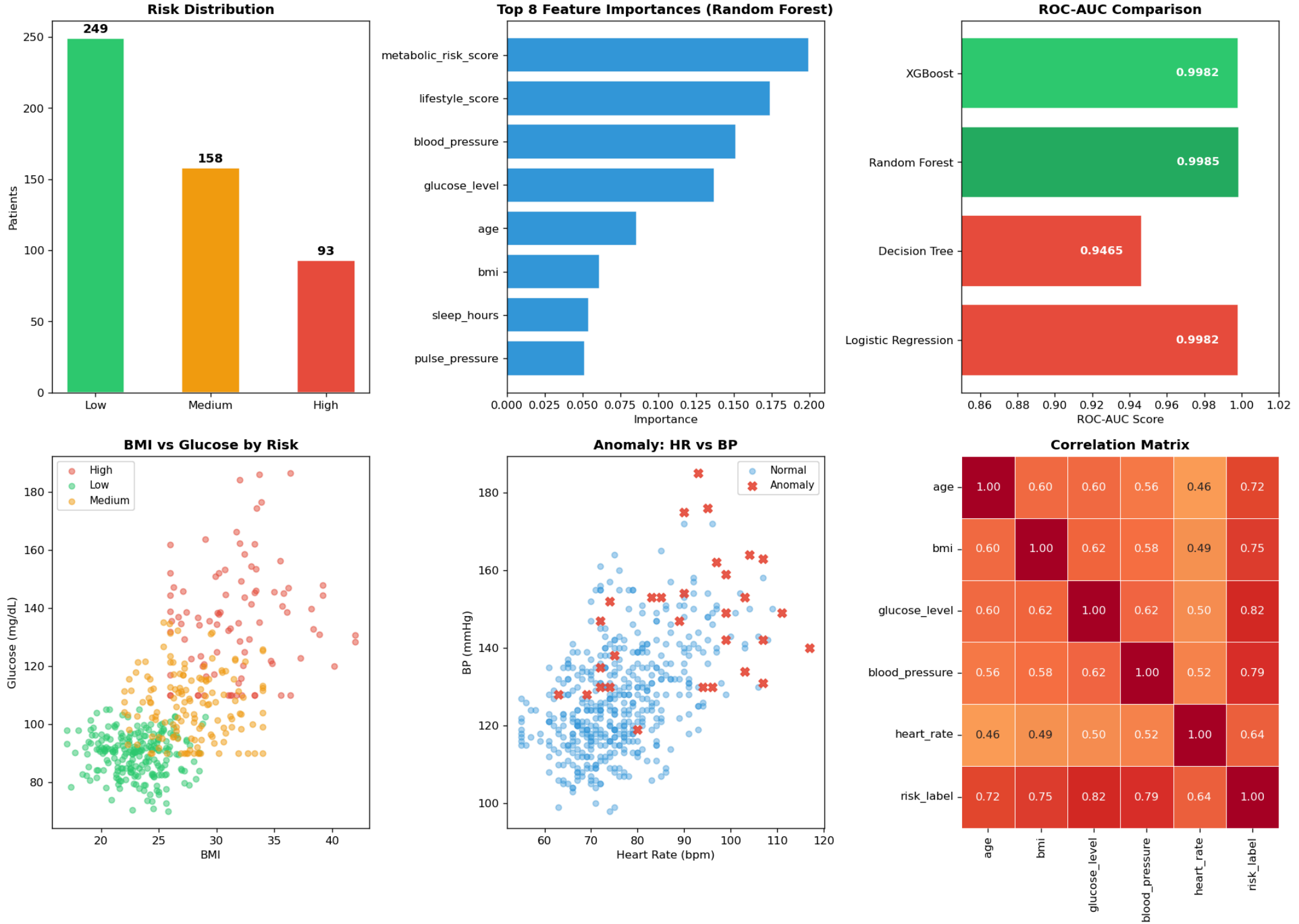

Feature Distributions by Risk Level
Age
age
Bmi
bmi
Glucose Level
glucose_level
Blood Pressure
blood_pressure
Heart Rate
heart_rate
Sleep Hours
sleep_hours
Low
Medium
High
Risk Level


Correlation Matrix
age
blood_pressure
heart_rate
bmi
glucose_level
sleep_hours
risk_label
1.00 0.56 0.46 0.60 0.60 -0.51 0.72
0.56 1.00 0.52 0.58 0.62 -0.54 0.79
0.46 0.52 1.00 0.49 0.50 -0.40 0.64
0.60 0.58 0.49 1.00 0.62 -0.54 0.75
0.60 0.62 0.50 0.62 1.00 -0.61 0.82
-0.51 -0.54 -0.40 -0.54 -0.61 1.00 -0.73
0.72 0.79 0.64 0.75 0.82 -0.73 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Model Performance Comparison
Accuracy
Precision
Recall
ROC-AUC
Score
1.0
0.9
0.8
0.7
0.6
0.5
Random Forest
Logistic Regression
XGBoost
Decision Tree


Feature Importance — Random Forest
metabolic_risk_score
lifestyle_score
blood_pressure
glucose_level
age
bmi
sleep_hours
pulse_pressure
heart_rate
activity_level_High
activity_level_Low
smoking_status_Never
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Importance Score


Anomaly Detection Visualization
Isolation Forest: HR vs BP
Normal
Anomaly
Blood Pressure (mmHg)
Heart Rate (bpm)
Rule-Based: Dangerous Combination Flag
Normal combination
Dangerous combo
BMI=30
Glucose=130
Glucose Level (mg/dL)
BMI

Digital Health Twin — Risk Prediction Dashboard